

**ADDIS ABABA UNIVERSITY,
FACULTY OF INFORMATICS**



Department Computer Science

**Entrance examination for the Masters of Science
Programme in Computer Science**

Date: June 30, 2004
Time allowed: 3 hours

Full Name : _____

Grade Obtained : _____

Important Note

- **Write your name only on this page. Writing your name on the other pages will automatically disqualify you from this exam.**
- **Check that you have 16 different pages (including this cover page).**
- **Any person found cheating will automatically fail the entrance exam**
- **Use the back of your question papers as rough papers. If you have used all the backs of your papers you can request for rough papers. Using other non authorized papers will be considered as cheating.**
- Answer neatly to all the questions on the space provided on the question paper
- Write short, and grammatically and logically correct sentences

Part I: Computer organization and assembly (7 points total)

1. (One point) What is the importance of having a hierarchy of memory instead of a single memory in the computer architecture?
2. (One point) What is Microprogramming? What are the advantages and drawbacks of microprogramming compared to hardwired programming?
3. (One point) Explain briefly how the buses are used for the communication between several components of a computer (Example: CPU and memory)

4. (4 points) Write two assembly language subroutines for 8086 architecture that will respectively encrypt and decrypt data located at memory address \$2004 using the famous Cesar's encryption algorithm. Cesar's encryption algorithm consists of shifting every letter two positions to the right in the alphabet. For decryption, the encrypted letter will be shifted by two positions to the left in order to obtain the original letter. For the special cases of 'y' and 'z', they will be encrypted as 'a' and 'b' respectively. For example: 'a' will be encrypted to 'c', 'b' to 'd'...'z' to 'b' etc. Therefore, the word "hi" will be encrypted as "jk".

Part II: Programming (17 points total)

[You can use a programming language of your preference.]

1.
 - a. Define a recursive function.[**1 points**]

 - b. Write a recursive function that computes the greatest common divisor (GCD) of two positive integers.[**5 points**]

2.
 - a. Write a function that accepts a positive integer and displays the digits of the number in reverse order.[**3 points**]

b. Analyze your subprogram for big O.[**1 points**]

c. Write a detailed algorithm of quick sort algorithm.[**8 points**]

Part III: Software Engineering (14 points total)

1.

a. Define Formal Software Specification. **(1 pt.)**

b. Though the idea of Formal Specification to software is novel, it is not widely used in the software industry. Describe the reasons behind this.
(2 pts.)

2.

a. Define software system design. **(1 pt.)**

b. Compare and contrast function-oriented and object-oriented design methods. (maximum of four lines for each design method). **(3 pts.)**

- c. What are the three most important design goals during the design of an interactive system which is to be used by non-computer literate users? **(2 pts.)**

3.

- a. Define regression testing. (1 pt.)

- b. How are test cases generated during regression testing? (2 pts.)

- c. What coverage criteria shall be used for regression testing? (2 pts.)

Part IV: Database Systems (14 points total)

1. (2 points) Why is indexing very important in databases?
2. (2 points) Explain what is a view in relational databases and for what purpose it is used?
3. (2 points) How can a Database administrator manage the security of the database using SQL commands?
4. A computer dealer maintains the database of all types of computers and peripherals that he sells to customers.
The following are the database schema and consists of four tables.
Product (manufacturer, model, type)
PC (model, speed, ram, hard-disk, removable-disk, price)
Laptop (model, speed, ram, hard-disk, screen, price)
Printer (model, color, type, price)

The **Product** table gives the manufacturer, model number and type (PC, Laptop, or Printer) of various products. Assume that model numbers are unique over all manufacturers and product types. The **PC** table gives for each model number that is a PC the speed (of the processor, in megahertz), the amount of RAM (in megabytes), the size of the hard disk (in gigabytes), the speed and type of the removable disk (CD or DVD), and the price. The **Laptop** table is similar, except that the screen size (in inches) is recorded in place of information about the removable disk. The **Printer** table records for each printer model whether the printer produces color output (true, if so), the process type (laser, ink-jet, or bubble), and the price. Using the above information answer the queries below:

i) (1 point) Write an SQL statement that give the manufacturer and speed of Laptops with hard disk of at least thirty giga bytes.

ii) (2.5 points) Write an SQL statement that finds the manufacturer of the color printer with the lowest price.

iii) (1.5 points) Write an SQL statement that find, for each different speed the average price of a PC.

5. (3 points) A database is being constructed to keep track of the teams and games of EURO 2004. A team has a number of players. It is desired to keep track of the players participating in each game for each team that they played in a game and the result of the game.
Design an ER-schema diagram for this application. Stating any assumptions you make.

Part V: Operating system (14 points total)

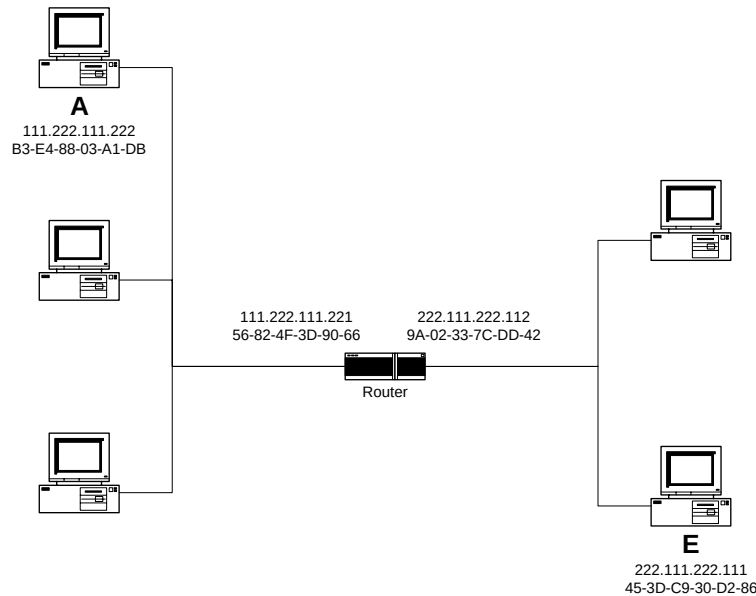
1. (3 points) Explain why a time-sharing operating system is not suitable for applications with hard real time requirements.
2. (2 points) How can processes communicate in a distributed system where there is no shared memory?
3. (4 points total) In modern operating systems, we can run programs with huge memory requirements even on computers with small amounts of RAM.
 - i) What techniques make this possible?
 - ii) So why buy large amounts of RAM at all?
4. (3 points total) Suppose a brand new kind of printer is invented this week. Even though this printer didn't exist when Windows 98 was released, we may still be able to configure a computer running Windows 98 so that it can print documents to this printer.
 - i) What change, if any, would you have to make to the system to allow it to print to this new printer?
 - ii) Explain how the operating system has been designed to run independently of device-specific details.
5. (2 points) You have probably had the experience of trying to read a file from a floppy disk and having the operating system tell you that the diskette is unreadable. What do you think is the most common reason for this failure?

Part VI: Networking (14 points total)

1. (2 points) Describe the advantages and the disadvantages of fiber optics cable compared to copper cables for computer communications?
2. (2 points) Explain what Ethernet is? How it solves the collision problem?
3. (4 points) Explain the difference between a *connection-oriented* network service and a *connectionless* service. Give an example of an application for which a connection-oriented service makes more sense, and an example of an application for which a connectionless service makes more sense. In both cases, briefly explain why.

4. (2 points) Explain the fundamental differences between a *hub* and a *switch*.

5. (4 points total) Consider the following simple network:



- i) (2 points) Write down an IP address for *each* interface at all hosts and routers in the network. The IP addresses for A and E are as given. You should assign IP addresses so that interfaces on the same network have the same network-part of their IP address. Indicate the number of bits in the network-part of this address.
- ii) (2 points) How do A, E and the router determine the IP addresses needed for an IP datagram moving from A to E?

Part VII: General questions on ICT in Ethiopia and abroad (20 points total)

Instruction: Attempt any two of the following five questions within the available spaces.

If you attempt more than two all of them will be discarded.

Each question is given 10 points as long as you follow the above instructions.

The following will be considered for the grade: Quality of language, Writing skills, Correctness and Quality of arguments and Neatness.

Your answers should have about 20 lines (average handwriting size).

- 1) What are Open Source Software (OSS) and proprietary software? Give two examples of OSS and two examples of proprietary software. What are the main differences between the two? Do you think OSS is good for Ethiopia? Explain.
- 2) Ethiopian audio-visual artists won a major battle this year? What is it? Do you think it has any relevance for Information and Communication Technology professionals? Explain.
- 3) Spams are becoming a major problem for internet users. Describe what spams are, what are the problems they cause and what solutions exist for individuals. Propose ideas on how spams can be eliminated (or at least highly limited) from the Internet in the future.
- 4) Describe the Weredanet and the SchoolNet project of the Ethiopian Government. What are the opportunities and the challenges ahead?
- 5) Microsoft is going to develop an Amharic version of Windows XP and Microsoft Office. There is already an Amharic version of Linux. Do you think that this evolution is very important for Ethiopia? Explain.

Answer in the following pages.

First chosen question (write just the number): _____

Answer below

Second chosen question (write just the number): _____

Answer below